

$$\begin{array}{c} \{\hat{\rho}_i\} \\ \vdots \\ \text{---} \boxed{\text{---}} \\ \vdots \end{array} \iff \mathbb{P} = \{p_i\}, \quad p_i = \text{tr}[(\hat{\rho}_i^{[0]} \otimes \hat{I}^{[1]})(\dots)] \\ \text{---} = \text{tr}[\hat{\rho}_i \text{tr}_1[(\dots)]]$$